# Problem 1:

1. **Characteristic Equation:** $r^2 - 5r + 6 = 0 \implies (r-2)(r-3) = 0.$

2. **Roots:** $r_1 = 2, r_2 = 3.$

3. **General Solution:** $F(n) = C_1(2^n) + C_2(3^n).$

4. **Initial Conditions:**
   - $n = 0 : C_1 + C_2 = 0 \implies C_1 = -C_2$
   - $n = 1 : 2C_1 + 3C_2 = 1 \implies -2C_2 + 3C_2 = 1 \implies C_2 = 1, C_1 = -1$

5. **Final Answer:** $F(n) = 3^n - 2^n$

# Problem 2:

1. **Homogeneous Part:** From Prob 1, $F_h(n) = C_1 2^n + C_2 3^n.$

2. **Particular Solution:** Guess $F_p(n) = An + B.$
   - Substitute: $(A(n+2) + B) = 5(A(n+1) + B) - 6(An + B) + 2n + 1$
   - Simplify: $An + (2A + B) = n(-A + 2) + (5A - B + 1)$
   - Solve: $A = -A + 2 \implies A = 1$; $2(1) + B = 5(1) - B + 1 \implies 2B = 4 \implies B = 2.$
   - $F_p(n) = n + 2.$

3. **General Solution:** $F(n) = C_1 2^n + C_2 3^n + n + 2.$

4. **Initial Conditions:**
   - $F(0) = 1 : C_1 + C_2 + 2 = 1 \implies C_1 + C_2 = -1$
   - $F(1) = 2 : 2C_1 + 3C_2 + 3 = 2 \implies 2C_1 + 3C_2 = -1$
   - Solving system: $C_2 = 1, C_1 = -2.$

5. **Final Answer:** $F(n) = 3^n - 2 \cdot 2^n + n + 2$

# Problem 3:

1. **Characteristic Equation:** $r^3 - 8r^2 + 17r - 10 = 0.$

2. **Roots:** By inspection $r = 1$ is a root. $(r-1)(r^2-7r+10) = (r-1)(r-2)(r-5) = 0.$

3. **Roots:** $r_1 = 1, r_2 = 2, r_3 = 5.$

4. **General Solution:** $F(n) = C_1 + C_2 2^n + C_3 5^n.$

5. **Initial Conditions:**
   - $n = 0 : C_1 + C_2 + C_3 = 4$

- $n = 1 : C_1 + 2C_2 + 5C_3 = 11$
- $n = 2 : C_1 + 4C_2 + 25C_3 = 49$
- Solving system: $C_3 = 2, C_2 = -1, C_1 = 3$.

6. **Final Answer:** $F(n) = 3 - 2^n + 2 \cdot 5^n$

# Problem 4:

1. **Parameters:** $a = 2, b = 4, f(n) = n + 42 \implies d = 1/2$.

2. **Comparison:** Compare $\log_b a$ and $d$.

   - $\log_4 2 = 1/2$.
   - $d = 1/2$.

3. **Result:** $T(n) = O(\sqrt{n} \log n)$, since $\log_b a = d$ $(1/2 = 1/2)$,.

# Problem 5:

1. **Parameters:** $a = 1, b = 2, f(n) = \frac{1}{2}n^2 + n \implies d = 2$.

2. **Comparison:** Compare $\log_b a$ and $d$.

   - $\log_2 1 = 0$.
   - $d = 2$.

3. **Result:** $T(n) = O(n^2)$, since $\log_b a < d$ $(0 < 2)$.

# Problem 6:

1. **Fax:** The best case, for boxer $i$ with $n$ victories, is to win the boxer $b$ with $n - 1$ victories. So for a boxer with $n$ victories, he first of all should already won $n - 1$ fights, and he need the other guy with $n - 2$ fights. It seems like the Fibonacci:

   - $F[0] = 1, F[1] = 2, F[n] = F[n-1] + F[n-2]$. We're lucky enough, that $F[8]$ is exactly 55.

2. **Result:** 8 victories is the maximum amount that is reachable.

# Problem 11:

1. **What is it:** It's the cut version of DP by profile, instead $n = 3$ and $m = 10$

2. **Here's my full solution from 2024 for full version of the problem from CSES:**

```cpp
#pragma GCC optimize ("Ofast")
#include <bits/stdc++.h>
using namespace std;
using ll = long long;

#define Ferarri ios_base::sync_with_stdio(0); cin.tie(NULL); cout.tie(NULL);

ll dp[1001][11][(1 << 11)], mod = 1e9 + 7;

signed main() {
    Ferarri
    ll t = 1;
    while (t--) {
        ll n, m;
        cin >> m >> n;
        dp[0][0][0] = 1;
        for (ll i = 0; i < n; i++) {
            for (ll j = 0; j < m; j++) {
                for (ll mask = 0; mask < (1 << m); mask++) {
                    if (mask & (1 << j)) {
                        dp[i][j + 1][mask - (1 << j)] += dp[i][j][mask];
                        dp[i][j + 1][mask - (1 << j)] %= mod;
                    }
                    else {
                        dp[i][j + 1][mask + (1 << j)] += dp[i][j][mask];
                        dp[i][j + 1][mask + (1 << j)] %= mod;
                        if (j + 1 < m && !(mask & (1 << (j + 1)))) {
                            dp[i][j + 1][mask + (1 << (j + 1))] += dp[i][j][mask];
                            dp[i][j + 1][mask + (1 << (j + 1))] %= mod;
                        }
                    }
                }
            }
            for (ll mask = 0; mask < (1 << m); mask++) {
                dp[i + 1][0][mask] = dp[i][m][mask];
            }
        }
        cout << dp[n][0][0];
    }
}
```

3. **Result:** As we can see, if we type for n = 3 and m = 10, the answer will be: 571